

# AI Prompt Optimizer - Professional Startup PRD

## AI Prompt Optimizer (PromptFix AI) Professional Product Requirements Document

Version: 1.0 MVP Owner: AI Builder Lab Executive Summary: AI Prompt Optimizer is a SaaS-style AI application that evaluates prompt quality and transforms weak prompts into professional-grade prompts. The product is designed to help users achieve better outputs from AI models while serving as a foundation for future AI Builder Lab products.

### Business Context

The rise of ChatGPT, Gemini, Claude, Cursor, and AI agents has created a demand for prompt engineering. Most users lack the skills required to structure prompts effectively. This product reduces that gap through automated prompt analysis and optimization.

### Problem Statement

Users frequently write vague prompts resulting in poor AI outputs. Common problems include missing context, unclear objectives, lack of constraints, undefined output formats, and absent audience definitions.

### Product Vision

Become the Grammarly for AI prompts by providing analysis, scoring, optimization, and education about prompt quality.

### Goals and Objectives

Primary Goals: • Improve prompt quality • Teach prompt engineering • Create YouTube-friendly project content • Build a reusable AI platform foundation Success Metrics: • Response time under 5 seconds • 95%+ valid AI responses • High user satisfaction

### Target Users

Primary Users: • Students • Developers • AI Beginners • Content Creators Secondary Users: • Marketers • Researchers • Freelancers • Prompt Engineers

### Competitive Analysis

Competitors: • Generic AI chatbots • Prompt marketplaces • Prompt engineering guides Differentiation: • Automated scoring • Structured feedback • Optimization engine • Educational value

### User Stories

As a user, I want to submit a prompt so that I can improve its quality. As a creator, I want prompt suggestions so that I can generate better content. As a developer, I want structured prompts so that AI produces more reliable outputs.

### Functional Requirements

FR-1 Prompt Submission FR-2 Prompt Analysis FR-3 Scoring Engine FR-4 Issue Detection FR-5 Suggestion Engine FR-6 Optimized Prompt Generation FR-7 Copy-to-Clipboard FR-8 Error

Handling

## Non-Functional Requirements

Performance: • <5 second response time Reliability: • Graceful error handling Security: • Input validation • Environment variable protection • No prompt storage by default Maintainability: • Clean architecture • Type safety • Modular services

## Technical Architecture

Frontend: Next.js 16, TypeScript, Tailwind CSS Backend: Python 3.13, FastAPI, Pydantic AI Layer: Gemini 2.5 Flash Deployment: Vercel + Railway Architecture: User → Frontend → FastAPI → Gemini API

## Data Flow

1. User submits prompt. 2. Frontend validates input. 3. FastAPI receives request. 4. Gemini evaluates prompt. 5. Structured JSON response generated. 6. Frontend displays score and optimized prompt.

## API Specification

POST /api/v1/optimize Request: { "prompt": "Write a blog about AI" } Response: { "overall\_score": 72, "issues": [...], "suggestions": [...], "optimized\_prompt": "..." }

## AI Design

Prompt Evaluation Dimensions: • Clarity • Context • Specificity • Constraints • Output Format AI returns structured JSON only.

## Frontend UX Requirements

Pages: • Landing Page • Results Section Components: • Prompt Form • Score Card • Suggestions List • Optimized Prompt Viewer • Copy Button

## Security & Risk Assessment

Risks: • AI hallucinations • Invalid JSON • API outages Mitigations: • Schema validation • Retry logic • Error boundaries

## Testing Strategy

Backend: • Unit tests • Integration tests Frontend: • Component tests • Form validation tests Manual: • End-to-end testing

## Deployment Strategy

Development: Local Docker environment Staging: Railway Preview Production: Vercel Frontend Railway Backend Monitoring: Logging and uptime monitoring

## Product Roadmap

V1: Prompt Optimization V2: Templates Multi-model support V3: Accounts Prompt history V4: Prompt Coach Prompt A/B Testing Team Collaboration

# YouTube Content Strategy

Video 1: I Built an AI That Fixes Bad Prompts Video 2: Build Tutorial Video 3: Testing Bad Prompts Against AI The project exists to support content creation and channel growth while teaching practical AI engineering.

## Conclusion

AI Prompt Optimizer is intentionally scoped to be achievable, educational, portfolio-worthy, and suitable for YouTube content production. It introduces a professional architecture using Next.js, FastAPI, and Gemini while remaining small enough to complete quickly.